SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 3 – SHORT ANSWER TEST

Course Code:	CSA0309	**Course Title:**	Data Structures
CO Assessed:	CO5 – Manipulation (BL3, BL4)	**Assessment:**	Assessment Tool 3 – Short Answer Test
Weightage:	30% of CIA	**Total Marks:**	100
CO2 Statement:	*Construct MST using shortest path algorithms and traverse the graph to model and analyse real-world problem.*		
Student Name:	A·Sathwika Reddy	**Reg. No.:**	192525274
Section / Batch:	2025	**Date:**	02/09/2026

Code of Conduct

I A·Sathwika Reddy – 192525274 *(Name / Reg No)*
certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.
Signature of the Student: A·Sathwika Reddy

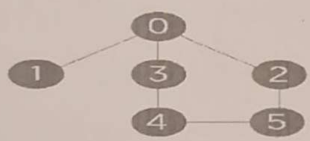
Instructions

- This test contains 10 short answer test questions. Total: 100 Marks.
- When a student answer using an Analytical problem-solving, in evaluation the answer should be with following five requirements: Understanding of problem, select appropriate data structure, Design the solution, analyze, Clarity of representation.
- No negative marking. Unanswered questions carry zero marks.
- No DTP printouts or electronic devices permitted. They should provide written materials.

Section / Topic	Focus	Q Nos	Marks
Minimum Spanning Tree	Kruskal's Algorithm	1	20
Graph Sorting	Topological Sort	2	20
Shortest Path Algorithm	MST & Dijkstra's Algorithm	3	20
Shortest Path Algorithm	Dijkstra's Algorithm	4	20
Minimum Spanning Tree	Prim's or Kruskal's Algorithm	5	20
	GRAND TOTAL:	100 Marks	

Annexure A – Short Answer Test

1. Perform the following operations in the Singly Linked List for the following input.
   Input: 22-> 77-> 55-> 88->
   i. Insert a new-node 110 at the end of the list.
   ii. Insert a new-node 101 at the position 4 of the list.
   iii. Delete a node at the beginning of the single linked list
   iv. Delete a node at the position 3 of the single linked list.
2. Rotations in each step of insertion. 1,2,3,4,5,6,7,8,9,10 (), 4) Push (90), 5) Push (36), 6) Push (11), 7) Push (88), 8) Pop (), 9) Pop (). Draw the diagram of the stack and illustrate the above operations and identify where the top is?
3. Convert the following infix into a postfix expression. → B - (C/D + (E % F* G) / H) * J
4. Evaluate the given postfix expression. 53+62/*35*+
5. Construct a B-Tree of Order 3 by inserting numbers 34,26,45,12,87,56,78,22,23,24,65,85,95.
6. Construct a 2-3-4 Tree by inserting the following sequence of numbers 45, 36, 76, 23, 89, 115, 98, 39, 41, 56, 69, 48.
7. Build a max heap for the following numbers - 20, 35, 23, 22, 4, 45, 21, 5, 42 and 19. At each stage of heap tree construction, ensure heap properties are satisfied.
8. Perform the operations onto splay tree (draw after each operation): insert (3), insert (9), insert (12), insert (55), insert (1), insert (3), delete (3), delete (9), delete (60), insert (3), search (12).
9. Describe the algorithm to sort the elements 170, 45, 75, 90, 802, 24, 2, 66 using Radix sort. Explain with each phase?
10. Consider the following graph & construct BFS.



Graph with six nodes

Rubrics for Calculation

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Conceptual Understanding	30	Demonstrates complete and accurate understanding of concepts	Good understanding with minor gaps	Basic understanding	Limited understanding
Accuracy of Answer	25	Answers are completely correct and relevant	Mostly correct with minor errors	Partially correct	Mostly incorrect
Problem-Solving & Reasoning	20	Provides logical reasoning and appropriate steps	Good reasoning with minor gaps	Basic reasoning	Lacks logical reasoning
Clarity & Relevance	15	Answers are precise, clear, concise, and directly address the question	Mostly clear and relevant	Somewhat unclear or incomplete	Irrelevant or unclear answers
Time Management & CompletionS	10	Completes all questions accurately within the time	Completes most questions on time	Requires additional time	Unable to complete the test

SFaculty In-charge	Course Coordinator	Program Director
Signature:	Signature: ____________	Signature: ____________
Date: ____________	Date: ____________	Date: ____________

Q1) Singly Linked list operations

22 → 77 → 55 → 88 → NULL

The required operations are insertion at end, insertion at position 4, deletion at begining, and deletion at position 3.

i, Insert 110 at the end

Traverse the list until the last node 88 is reached. Create new node containing 110 and make 88 point to it.

Before

22 → 77 → 55 → 88 → NULL

After

22 → 77 → 55 → 88 → 110 → NULL

ii, Insert 101 at position 4

position	Element
1	22
2	77
3	55
4	88
5	110

After insertion

22 → 77 → 55 → 101 → 88 → 110 → NULL

iii, Deletion a node at the beginning

The first node containing 22 is removed.

Before

22 → 77 → 55 → 101 → 88 → 110 → NULL

After

77 → 55 → 101 → 88 → 110 → NULL

The head is now pointing to 77.

iv, Delete node at position 3

current list

77 → 55 → 101 → 88 → 101 → NULL

Final list:-

77 → 55 → 88 → 110 → NULL

Result :-

All four linked-list operations are successfully performed.

2) The given operations are push(90), push(36), push(11), push(88), pop(), and pop().

* A Stack follow the LIFO (Last In, First out) principle. The element inserted last is removed first.

push (90)

| 90 | → TOP

push (36)

| 36 | → TOP
| 90 |

push (11)

| 11 | → TOP
| 36 |
| 90 |

push (88)

| 88 | → TOP
| 11 |
| 36 |
| 90 |

pop()

| 11 | ← TOP
| 36 |
| 90 |

pop()

| 36 | ← Top
| 90 |

Therefore

* first popped element = 88
* second popped element = 11
* Final stack = 90, 36
* Top = 36

Result:- Stack operations are successfully performed using LIFO.

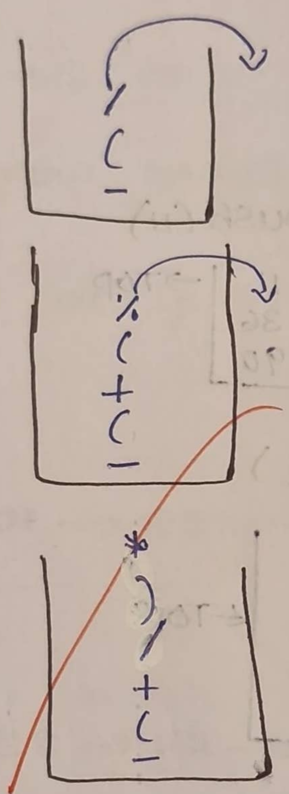
3) Infix to Postfix conversion

Given infix expression:

B − (C/D + (E%F * G)/H) * J

I/P

Stack



O/P

BCD/EF%G*H/+J*−

B
−
(
C
/
(
E
%
F
*
G
)
/
H
)
*
J

4) Given postfix expression

53 + 62 / * 35 * +

postfix expressions are evaluated using Stack.

Symbol	operation	stack
5	push	5
3	push	5, 3
+	5+3	8
6	push	8, 6
2	push	8, 6, 2
/	6/2	8, 3
*	8 x 3	24
3	push	24, 3
5	push	24, 3, 5
*	3x5	24, 15
+	24 + 15	39

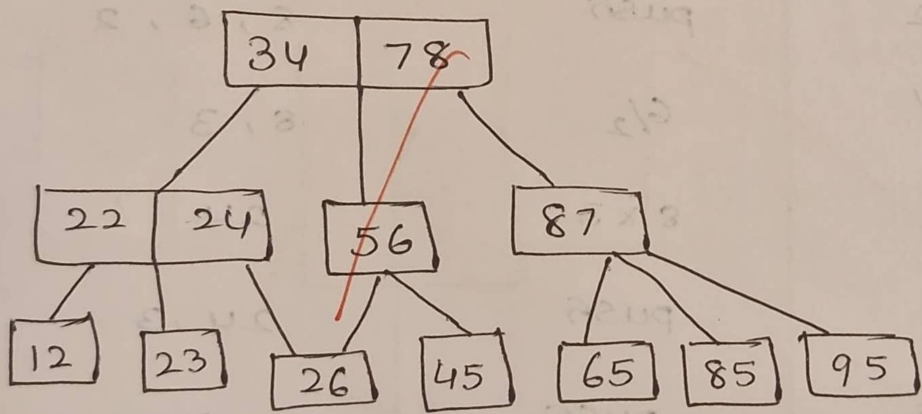
Result :- The given postfix expression evaluates to 39.

5) 34, 26, 45, 12, 87, 56, 78, 22, 23, 24, 65, 85, 95

* For a B-Tree of order 3, a node can contain a maximum of 2 keys and 3 children.

* When a node becomes full during insertion it is split and the middle key is promoted to parent

* After inserting all the given elements & performing the required splits, the tree can be represented as:

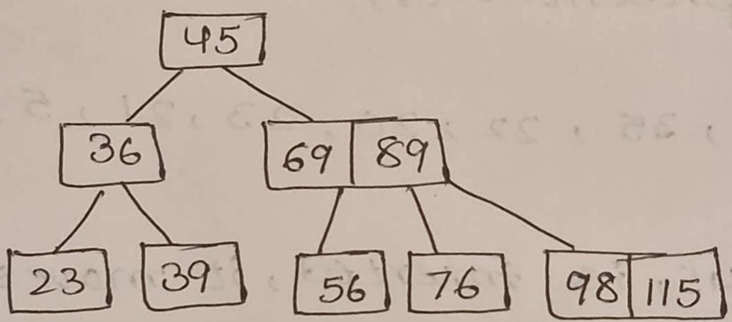


* The keys within each node are maintained in sorted order. All leaf nodes remain at the same level, which is an important property of a B-Tree.

6) Given sequence

45, 36, 76, 23, 89, 115, 98, 39, 41, 56, 69, 48

* A 2-3-4 Tree is balanced search Tree in which a node can contain 1, 2 or 3 keys
* A node containing three keys is split when another insertion would make it overflow
* After performing the insertions & necessary splits, the tree is represented as:



* The tree remains balanced because all leaf nodes occur at the same level.

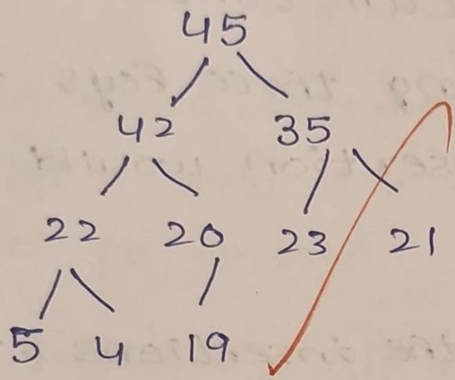
Result:- The given elements are inserted successfully while maintaining the properties of a 2-3-4 Tree.

7> Max Heap

Elements:

20, 35, 23, 22, 4, 45, 21, 5, 42, 19

Final Max Heap:



Array Representation

45, 42, 35, 22, 20, 23, 21, 5, 4, 19

* When 45, is inserted, it moves upward because it is greater than parent nodes. Similarly 42 is moved upward until the max-heap property is satisfied.

8) # Splay Tree

Operations include, insertion, deletion and search.

<u>Main rule</u> : Every accessed / inserted node is splayed (rotated) to the root.

Insert (3)

3

Insert (9) 9 is splayed to the root

```
  9
 /
3
```

Insert (12)

```
    12
   /
  9
 /
3
```

* After inserting 55, it becomes the root after splaying.
* After inserting ~~3~~ 1, newly inserted node 1 is splayed to root.

Delete (3) removes node 3

Delete (9) removes node 9

Delete (60) has no effect

Finally root = 12

9) Radix Sort

Given

170, 45, 75, 90, 802, 24, 2, 66

* Radix sort sorts numbers digit by digit Starting from least significant digit

0	1	2	3	4	5	6	7	8	9
90 170		2 802		24	75 45	66			

170, 90, 802, 2, 24, 45, 75, 66

0	1	2	3	4	5	6	7	8	9
2 802		24		45		66	75 170		90

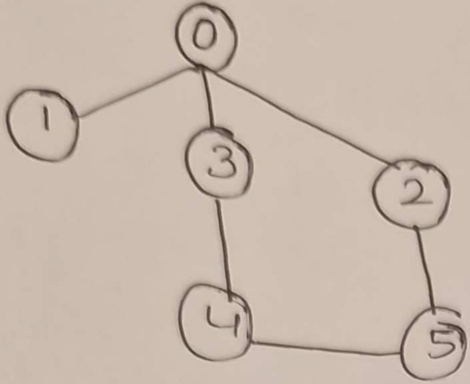
802, 2, 24, 45, 66, 75, 170, 90

0	1	2	3	4	5	6	7	8	9
90 75 66 45 24 2	175							802	

Sorted array.

2, 24, 45, 66, 75, 90, 175, 802

10>



0					

visited queue

1	3	2			

unvisited queue

0	1				

visited queue

3	2				

unvisited queue

0	1	3			

visited queue

2	4				

unvisited queue

0	1	3	2		

visited queue

4	5				

unvisited queue

Final correct order

0	1	3	2	4	5